

ClientIQ On-Premises Deployment Guide

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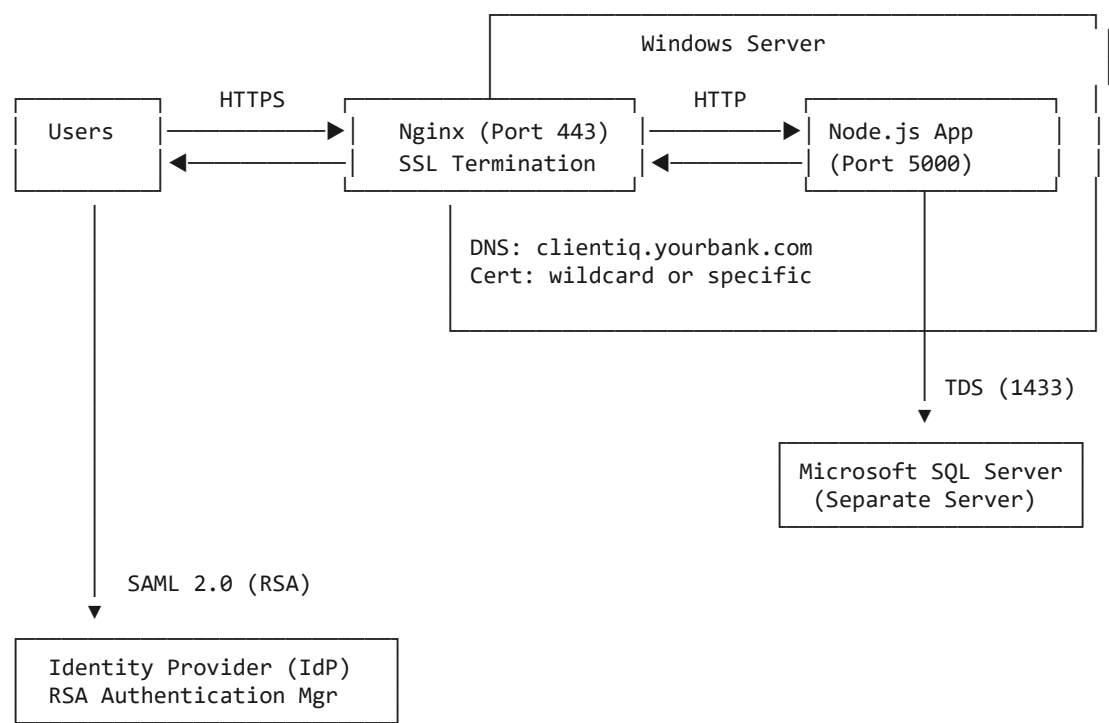
ClientIQ On-Premises Deployment Guide

Overview

This documentation provides comprehensive instructions for deploying the ClientIQ Banking Client 360 application in an on-premises Windows Server environment. The deployment architecture uses:

- **Windows Server 2019/2022** - Hosting the Node.js application
- **Microsoft SQL Server 2019+** - Backend database
- **Nginx** - Reverse proxy with SSL termination
- **SAML 2.0** - Enterprise authentication via RSA Identity Provider

Architecture Diagram



Documentation Index

Document	Audience	Description
01-WINDOWS-SERVER-SETUP.md	System Admin	Windows Server and Node.js configuration (Production)
02-NGINX-REVERSE-PROXY.md	System Admin	Nginx installation and reverse proxy setup
03-SSL-CERTIFICATE-SETUP.md	System Admin	SSL/TLS certificates and DNS mapping
04-SAML-CONFIGURATION.md	Security Team	SAML 2.0 authentication with RSA IdP
05-DATABASE-DESIGN.md	Data Engineer	Database schema, relationships, and data model
06-MSSQL-DBA-GUIDE.md	Database Admin	SQL Server setup, security, and maintenance
07-ENVIRONMENT-VARIABLES.md	All	Complete environment configuration reference
08-TROUBLESHOOTING.md	All	Common issues and solutions
09-DEV-TEST-SERVER-SETUP.md	System Admin	Development/Test server deployment guide

Deployment Checklist

Pre-Deployment

- ☐ Windows Server 2019/2022 provisioned with minimum 8GB RAM
- ☐ SQL Server 2019+ instance available with network access
- ☐ DNS record created (e.g., `clientiq.yourbank.com`)
- ☐ SSL certificate obtained (wildcard or specific hostname)
- ☐ RSA IdP configured with SAML 2.0 endpoints
- ☐ Network firewall rules configured (443 inbound, 1433 to SQL Server)
- ☐ Service account credentials for SQL Server
- ☐ Node.js LTS (v20+) installer downloaded

Deployment Steps

1. **Windows Server Setup** (Document 01)

- Install Node.js
- Create application directory
- Configure Windows services
- Set up logging

2. **Nginx Configuration** (Document 02)

- Install Nginx for Windows
- Configure reverse proxy
- Set up health checks
- Configure access logging

3. **SSL/DNS Setup** (Document 03)

- Install SSL certificate
- Configure DNS mapping
- Enable HTTPS redirection
- Test certificate chain

4. **SAML Configuration** (Document 04)

- Exchange metadata with IdP
- Configure attribute mapping
- Set up role synchronization
- Test SSO flow

5. **Database Setup** (Document 06)

- Create database and schema
- Import reference data
- Configure security
- Set up backups

6. **Application Deployment**

- Extract application package
- Configure environment variables
- Start application service
- Verify health endpoints

Post-Deployment Verification

- ☐ HTTPS accessible at `https://clientiq.yourbank.com`
- ☐ SSL certificate valid and trusted
- ☐ SAML login redirects to IdP
- ☐ User can authenticate and access application
- ☐ Database queries execute successfully
- ☐ Session management working
- ☐ Logs are being captured

System Requirements

Windows Server

Component	Minimum	Recommended
OS	Windows Server 2019	Windows Server 2022
CPU	4 cores	8 cores
RAM	8 GB	16 GB
Disk	50 GB SSD	100 GB SSD
Network	1 Gbps	10 Gbps

SQL Server

Component	Minimum	Recommended
Version	SQL Server 2019 Standard	SQL Server 2022 Enterprise
CPU	4 cores	8+ cores
RAM	16 GB	32+ GB
Disk	100 GB SSD	500+ GB SSD

Network Requirements

Port	Protocol	Direction	Purpose
443	TCP	Inbound	HTTPS traffic
80	TCP	Inbound	HTTP redirect (optional)
5000	TCP	Internal	Node.js application
1433	TCP	Outbound	SQL Server connection

Support Contacts

Role	Contact	Escalation
Application Support	app-support@yourbank.com	L1
Database Administration	dba-team@yourbank.com	L2
Security/SSO	security-team@yourbank.com	L2
Infrastructure	infra-team@yourbank.com	L3

Version History

Version	Date	Author	Changes
2.0.0	December 2025	ClientIQ Team	Initial on-prem documentation